TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	0.0 ± 0.00
Correlation Stability	82.0 ± 0.00
Distribution Consistency	0.0 ± 0.00
Entropy Preservation	0.0 ± 0.00
Mean Z Score Drift	0.0 ± 0.00
Categorical Fidelity	29.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	0.0 ± 0.00
Coverage Score	0.0 ± 0.00
Feature Accuracy	0.0 ± 0.00
Novelty Score	0.0 ± 0.00

TabuLLM: AI Benchmark Report

Dataset Audit: heart_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	303
Columns	14
Numeric features	14
Categorical features	0
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	0.0 ± 0.00
Correlation Stability (*2)	82.0 ± 0.00
Distribution Consistency (*3)	0.0 ± 0.00
Entropy Preservation (*4)	0.0 ± 0.00
Mean Z Score Drift (*5)	0.0 ± 0.00
Categorical Fidelity (*6)	29.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00
Data Density (*8)	0.0 ± 0.00
Coverage Score (*9)	0.0 ± 0.00
Feature Accuracy (*10)	0.0 ± 0.00
Novelty Score (*11)	0.0 ± 0.00

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.598360655737704	N/A	0.563934426229508	-0.03442622950819672	
	9		2		
F1	0.565074855402510	N/A	0.531101745761099	-0.03397310964141108	
	2				

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.557377049180327	N/A	0.522950819672131	-0.03442622950819671	
	9		1		
F1	0.568660979990255	N/A	0.530968563267476	-0.03769241672277819	
			9		

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.554098360655737	N/A	0.531147540983606	-0.02295081967213115	
	7		5	7	
F1	0.444220207143656	N/A	0.390745379690459	-0.05347482745319729	*
	84		5	5	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.547540983606557	N/A	0.547540983606557	0	
	3		3		
F1	0.389927060011082	N/A	0.389927060011082	0	
	7		7		

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.588524590163934	N/A	0.545901639344262	-0.04262295081967213	
	4		4		
F1	0.540674089326052	N/A	0.519007079739968	-0.02166700958608467	
	7			5	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6081967213114754	N/A	0.580327868852459	-0.02786885245901639	
F1	0.6070236204175147	N/A	0.5620042220725474	-0.04501939834496721	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5426229508196723	N/A	0.5540983606557377	0.011475409836065575	
F1	0.5198013329305267	N/A	0.5364410744931339	0.01663974156260724	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5344262295081967	N/A	0.5655737704918032	0.03114754098360654	
F1	0.5216185407173739	N/A	0.5331451642136888	0.011526623496314897	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5262295081967212	N/A	0.5	-0.02622950819672132	
F1	0.5285808546310123	N/A	0.5016529141141646	-0.026927940516847926	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5737704918032788	N/A	0.5918032786885246	0.01803278688524589	
F1	0.5309034754274821	N/A	0.5371627676846743	0.006259292257192161	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
2	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5
3	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6
4	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5
5	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4
6	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
7	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6
8	57.0	0.0	4.0	120.0	354.0	0.0	0.0	163.0	1.0	0.6
9	63.0	1.0	4.0	130.0	254.0	0.0	2.0	147.0	0.0	1.4
10	53.0	1.0	4.0	140.0	203.0	1.0	2.0	155.0	1.0	3.1
11	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
12	56.0	0.0	2.0	140.0	294.0	0.0	2.0	153.0	0.0	1.3
13	56.0	1.0	3.0	130.0	256.0	1.0	2.0	142.0	1.0	0.6
14	44.0	1.0	2.0	120.0	263.0	0.0	0.0	173.0	0.0	0.0
15	52.0	1.0	3.0	172.0	199.0	1.0	0.0	162.0	0.0	0.5
16	57.0	1.0	3.0	150.0	168.0	0.0	0.0	174.0	0.0	1.6
17	48.0	1.0	2.0	110.0	229.0	0.0	0.0	168.0	0.0	1.0
18	54.0	1.0	4.0	140.0	239.0	0.0	0.0	160.0	0.0	1.2
19	48.0	0.0	3.0	130.0	275.0	0.0	0.0	139.0	0.0	0.2
20	49.0	1.0	2.0	130.0	266.0	0.0	0.0	171.0	0.0	0.6

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	49.0	0.0	4.0	130.0	269.0	0.0	0.0	163.0	0.0	0.0
2	54.0	1.0	3.0	120.0	258.0	0.0	2.0	147.0	0.0	0.4
3	55.0	1.0	2.0	130.0	262.0	0.0	0.0	155.0	0.0	0.0
4	46.0	1.0	2.0	101.0	197.0	1.0	0.0	156.0	0.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
2	60.0	1.0	4.0	130.0	253.0	0.0	0.0	144.0	1.0	1.4
3	58.0	0.0	2.0	136.0	319.0	1.0	2.0	152.0	0.0	0.0
4	59.0	0.0	4.0	174.0	249.0	0.0	0.0	143.0	1.0	0.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
2	45.0	1.0	4.0	115.0	260.0	0.0	2.0	185.0	0.0	0.0
3	45.0	0.0	2.0	130.0	234.0	0.0	2.0	175.0	0.0	0.6
4	46.0	1.0	3.0	150.0	231.0	0.0	0.0	147.0	0.0	3.6

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
2	63.0	0.0	4.0	150.0	407.0	0.0	2.0	154.0	0.0	4.0
3	57.0	0.0	4.0	128.0	303.0	0.0	2.0	159.0	0.0	0.0
4	60.0	1.0	4.0	117.0	230.0	1.0	0.0	160.0	1.0	1.4

B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	50.0	0.0	2.0	120.0	244.0	0.0	0.0	162.0	0.0	1.1
2	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
3	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0
4	64.0	0.0	3.0	140.0	313.0	0.0	0.0	133.0	0.0	0.2

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	39.0	1.0	4.0	118.0	219.0	0.0	0.0	140.0	0.0	1.2
2	62.0	0.0	4.0	138.0	294.0	1.0	0.0	106.0	0.0	1.9
3	66.0	1.0	4.0	160.0	228.0	0.0	2.0	138.0	0.0	2.3
4	39.0	1.0	3.0	140.0	321.0	0.0	2.0	182.0	0.0	0.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	56.0	0.0	4.0	134.0	409.0	0.0	2.0	150.0	1.0	1.9
2	58.0	1.0	3.0	105.0	240.0	0.0	2.0	154.0	1.0	0.6
3	54.0	1.0	2.0	192.0	283.0	0.0	2.0	195.0	0.0	0.0
4	57.0	1.0	4.0	110.0	201.0	0.0	0.0	126.0	1.0	1.5

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	52.0	1.0	1.0	118.0	186.0	0.0	2.0	190.0	0.0	0.0
2	54.0	0.0	3.0	135.0	304.0	1.0	0.0	170.0	0.0	0.0
3	61.0	1.0	4.0	140.0	207.0	0.0	2.0	138.0	1.0	1.9
4	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	44.0	0.0	3.0	118.0	242.0	0.0	0.0	149.0	0.0	0.3
2	38.0	1.0	1.0	120.0	231.0	0.0	0.0	182.0	1.0	3.8
3	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
4	61.0	1.0	4.0	120.0	260.0	0.0	0.0	140.0	1.0	3.6

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
2	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
3	60.0	1.0	4.0	125.0	258.0	0.0	2.0	141.0	1.0	2.8
4	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6